

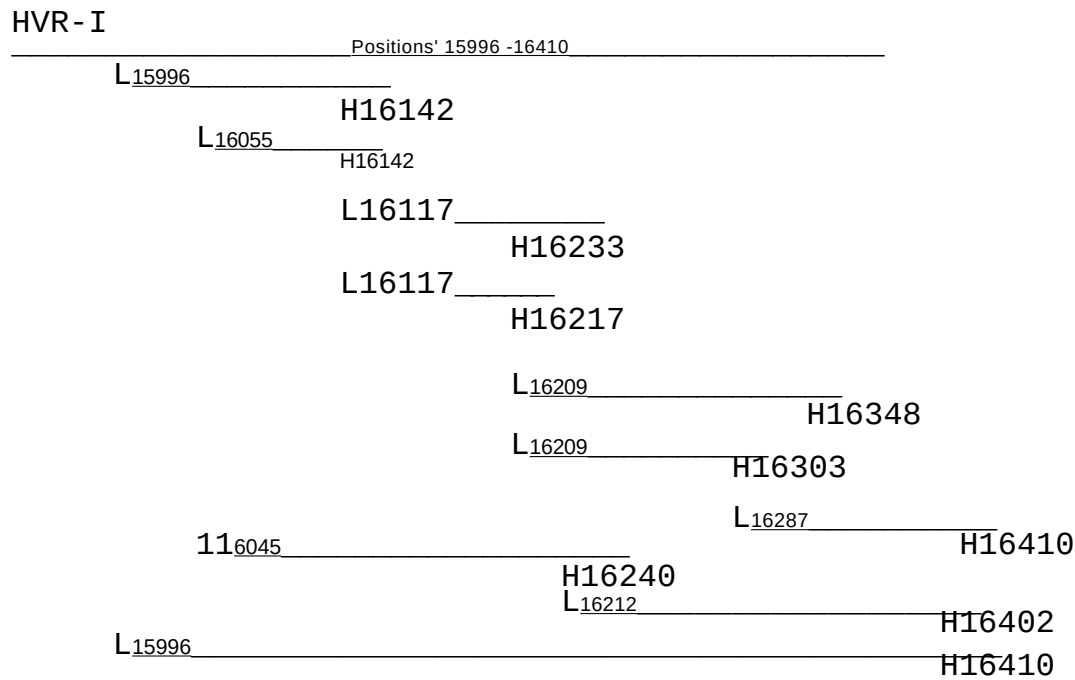


Figure 2: System of primers for HVR-I amplification and sequencing

Typing of 22 coding region SNPs (GenoCore22 reaction)

The assessment of mtDNA haplogroups based on HVR-I sequences was refined and/or confirmed by typing 22 haplogroup diagnostic SNPs in the coding region of the mtDNA genome.

SNaPshot minisequencing has been previously applied to type SNPs in the mitochondrial genome (Brandstatter et al., 2003; Brandstatter et al., 2006; Grignani et al., 2006; Niederstatter et al., 2006; Haak et al., 2010), on autosomal chromosomes (Grimes et al., 2001; Dixon et al., 2005; Sanchez et al., 2006; Bouakaze et al., 2007) and on the Y-chromosome (Sanchez et al., 2003; Sanchez et al., 2005; Brion et al., 2005a; Brion et al., 2005b; Onofri et al., 2006; Bouakaze et al., 2009). These studies include work on modern, forensic and ancient DNA and have demonstrated the robustness, reliability and sensitivity of the SNaPshot minisequencing approach (Bouakaze et al., 2009).

In general, coding region SNPs were characterised using a multiplex PCR followed by a multiplexed Single Base Extension (SBE) reaction (GenoCore22, Haak et al., 2010) using the SNaPshot minisequencing technique (Applied Biosystems). This approach consists of a preliminary multiplex PCR amplification, in the present

8994
6371
12705

11719
14766
7028
4580
8280
13928
13368
12612
11467
10550

case targeting 22 short fragments encompassing haplogroup-specific SNPs in the coding region (as defined in Behai' et al., 2007; Figure 3). A SBE reaction follows, where oligonucleotides designed to hybridise one base pair upstream of each SNP of interest are used as primers for the addition of a single dideoxyribonucleoside triphosphate (ddNTP). Each ddNTP is labelled with a different fluorochrom to allow detection of the SNPs after fragment length-based electrophoresis separation of the 22 amplicons.

The GenoCoRe22 reaction is particularly relevant to the study of ancient mtDNA for several reasons. First, because the GenoCoRe22 targets 22 SNPs in a single reaction it is a time and cost-efficient method that uses very small amounts of valuable aDNA extracts. Because it limits the number of PCRs necessary to type 22 SNPs, GenoCoRe22 also reduces the risk of introducing modern contamination in the typing reactions (as opposed to 22 single reactions). As the multiplex is designed to target short DNA fragments spanning sizes from 60 bp to 80 bp, it is suitable for highly degraded DNA. Finally, the GenoCoRe22 reaction is used here as an internal control. The concordance between the haplogroup assessments independently obtained from the analysis of HVR-I sequences and SNaPshot profiles provides a key means to authenticate the aDNA results. In addition, because of the hierarchical nature of its design, i.e., typing of SNPs diagnostic of mtDNA macro-haplogroups, haplogroups and sub-haplogroups, the GenoCore22 reaction allows the detection of contamination and/or DNA damage through the observation of superimposed and/or phylogenetically inconsistent profiles.

-L0
-1
-12
-6
-L3
-4
-11
-C
-D
-A
-N1
-W
-X
■R
■R
O
■HV
■H
-V
■B
-R9

Figure 3: Schematic representation of the human mitochondrial phylogeny indicating the positions of the Single Nucleotide Polymorphisms typed in the GenoCore22 reaction.

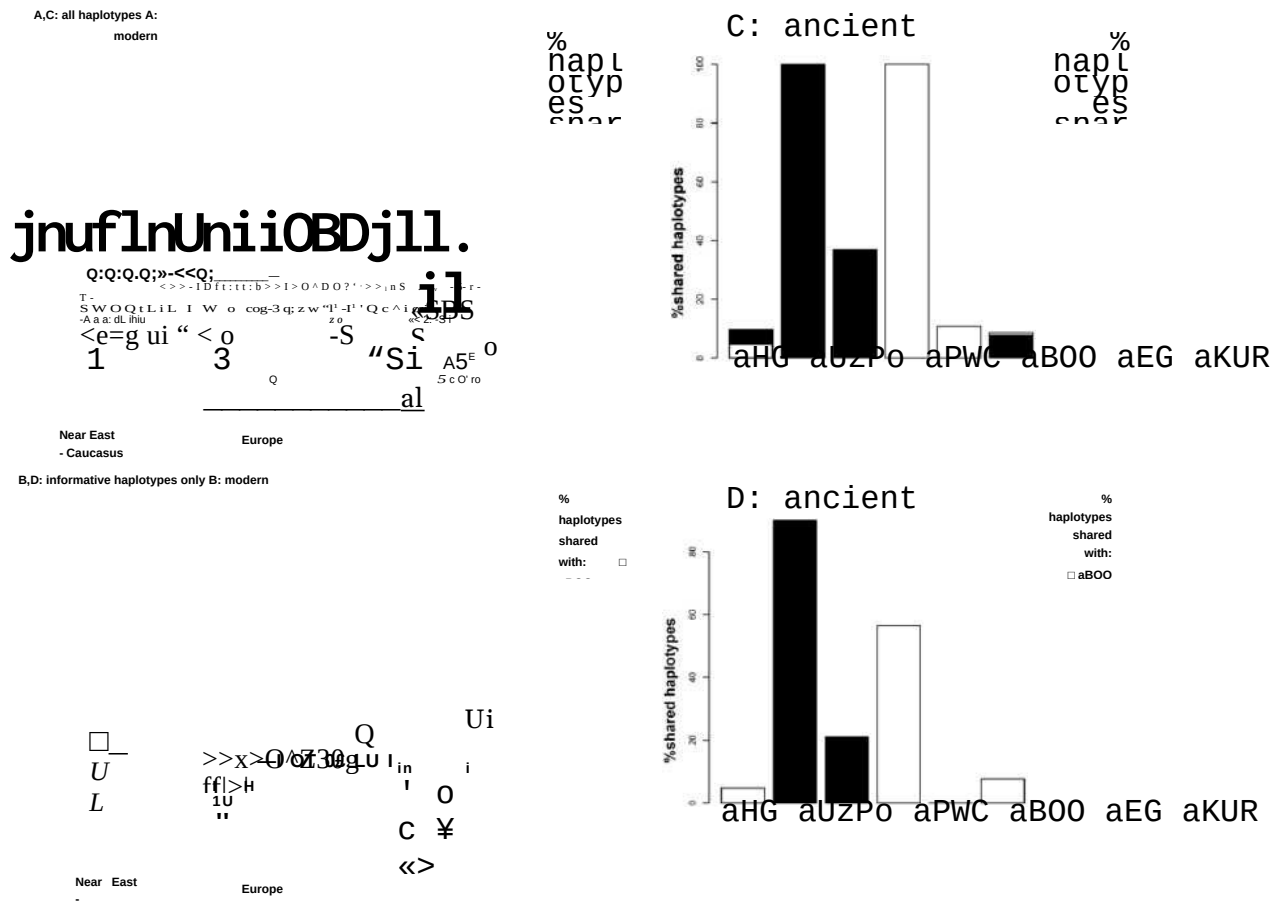


Figure 4. Percentages of Uznyi Oleni Ostrov/Popovo (black) and Bolshoy Oleni Ostrov (white) haplotypes matched in selected modern (A, B) and ancient (C, D) Eurasian population pools. The haplotype sharing analysis graph allows visualising the current distribution of the ancient haplotypes from Uznyi Oleni Ostrov/Popovo and Bolshoy Oleni Ostrov in modern populations from three main geographical regions: the Near-East/Caucasus, Europe, East Siberia (A, B). For each modern population, the number of occurrences of each haplotype from a given ancient population was divided by the modern population sample size (N = 500). The corresponding percentages of shared haplotypes are represented by a black bar for Uznyi Oleni Ostrov/Popovo haplotypes and by a white bar for Bolshoy Oleni Ostrov haplotypes. The same analyses were performed to compare the Uznyi Oleni Ostrov/Popovo and Bolshoy Oleni Ostrov haplotypes with haplotypes previously described in ancient populations (C, D). Figures A and C were constructed considering all haplotypes sequenced in the populations of Uznyi Oleni Ostrov/Popovo and Bolshoy Oleni Ostrov, whereas uninformative basal haplotypes were discarded in figures B and D. See Material and Methods for population abbreviations.